Math 118: Mathematical Methods of Data Theory

Lecture 1: Matrix algebra and matrix norms

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First: A Brief overview of this course

- **Tools:**
  - **Linear Algebra/ Functional Analysis:**
    - Matrix Decomposition: LU, QR, non-negative factorization.
    - Matrix Norms: Perturbation and Conditioning number.
    - Singular Values and Eigenvalues: Low rank approximation, PCA, Cheeger Inequality.
    - Tensors: tensor singular values.
  - **Optimization**
    - Linear Programming: Basic Solutions, Standard Form, Simplex and non-simplex methods.
    - Convex Optimization: Gradient descent, Newton's method, KKT conditions.
    - Integer Optimization: Branch-and-Bound.
    - Dynamic Programming: Optimal substructure, overlapping subproblems.

First: A Brief overview of this course

- Applications:
  - Solving Linear Systems and Least Squares, analyzing error.
  - Data Compression and denoising.
  - Detecting communities in social networks.
  - Classification: eg handwritten digits.
  - Latent Semantic Analysis.
  - PageRank.
  - Neural Networks.

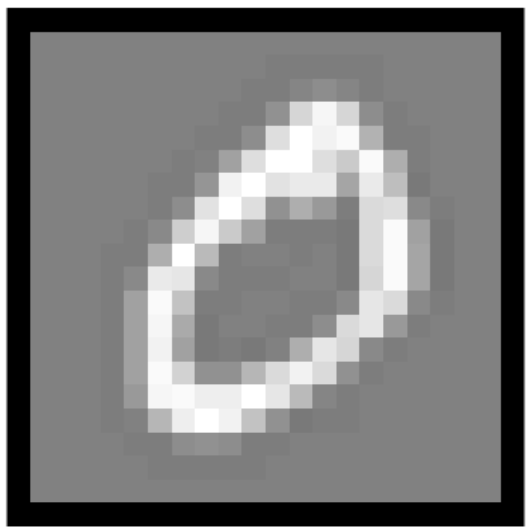
Matrices and vectors

- a **vector** is a 1-D array of numbers: $\boldsymbol{v} = \begin{bmatrix} 1 \\ 3 \\ -13.7 \\ 3.1415 \end{bmatrix}$
- If $\boldsymbol{v}$ has n entries we write $\boldsymbol{v} \in \mathbb{R}^n$.
- a **matrix** is a 2-D array of numbers: $A = \begin{bmatrix} 0 & 5 & -7 \\ 2 & 5.14 & 1.5 \end{bmatrix}$
- If A has m rows and n columns we write $A \in \mathbb{R}^{m \times n}$
- a **tensor** is an r-D array of numbers *more on this later*.

Constructing matrices and vectors from data

Example

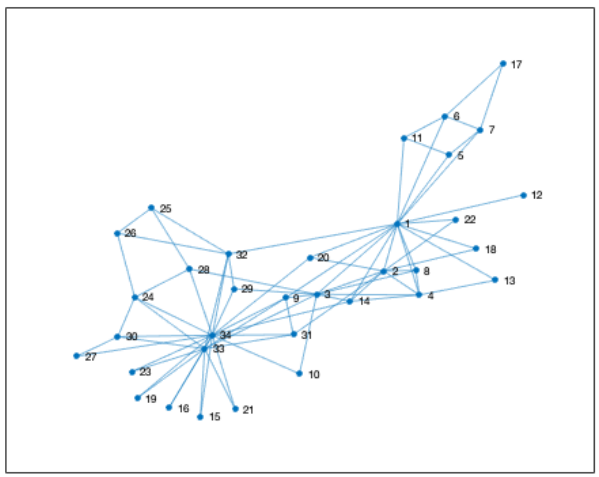
Grayscale images can be thought of as matrices:

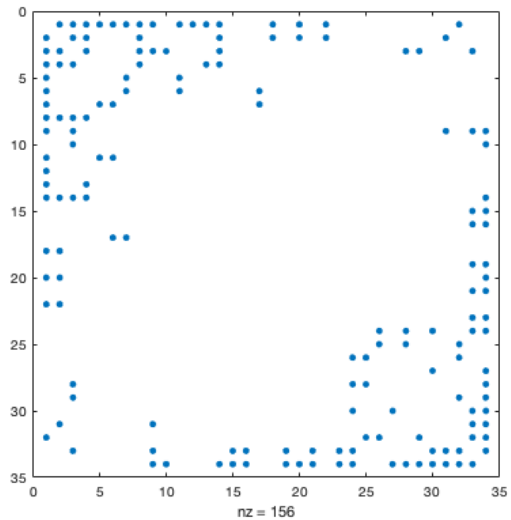


Constructing matrices and vectors from data

Example

Graphs can be studied via their adjacency matrix:

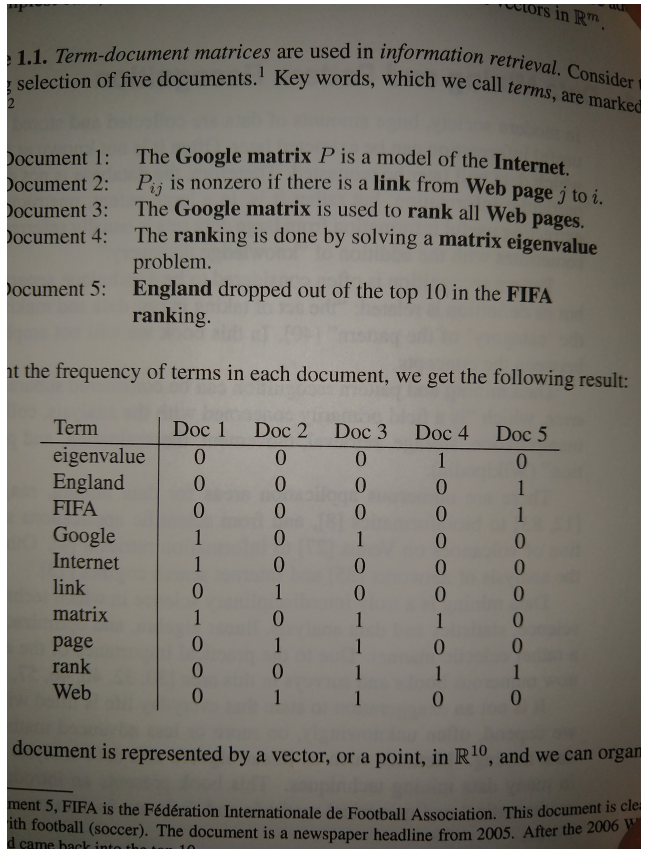




Constructing matrices and vectors from data

Example

Documents can be turned into vectors. Multiple documents can be collected into a matrix.



1.1. *Term-document matrices* are used in *information retrieval*. Consider
selection of five documents.[1] Key words, which we call *terms*, are marked

Document 1: The **Google matrix** P is a model of the **Internet**.
Document 2: P_{ij} is nonzero if there is a **link** from **Web page** j to i.
Document 3: The **Google matrix** is used to **rank** all **Web pages**.
Document 4: The **rank**ing is done by solving a **matrix eigenvalue** problem.
Document 5: **England** dropped out of the top 10 in the **FIFA rank**ing.

the frequency of terms in each document, we get the following result:

Term	Doc 1	Doc 2	Doc 3	Doc 4	Doc 5
eigenvalue	0	0	0	1	0
England	0	0	0	0	1
FIFA	0	0	0	0	1
Google	1	0	1	0	0
Internet	1	0	0	0	0
link	0	1	0	0	0
matrix	1	0	1	1	0
page	0	1	1	0	0
rank	0	0	1	1	1
Web	0	1	1	0	0

document is represented by a vector, or a point, in $\mathbb{R}^{10}$, and we can organ

ment 5, FIFA is the Fédération Internationale de Football Association. This document is cle
ith football (soccer). The document is a newspaper headline from 2005. After the 2006 W

Figure: From *Matrix Methods in Data Mining and Pattern Recognition* by Lars Elden

Matrix and vector multiplications

- For $A \in \mathbb{R}^{m \times n}$ and $\boldsymbol{v} \in \mathbb{R}^n$ can define a new vector $A\boldsymbol{v} \in \mathbb{R}^m$ as:

$$(A\boldsymbol{v})_i = \sum_{j=1}^{m} A_{ij} v_j$$

  Can think of A as a map/function from $\mathbb{R}^n$ to $\mathbb{R}^m$.

- For two vectors $\boldsymbol{u}$ and $\boldsymbol{v} \in \mathbb{R}^n$ can define inner product:

$$(\boldsymbol{u}, \boldsymbol{v}) = \boldsymbol{u}^\top \boldsymbol{v} = \sum_{i=1}^{n} u_i v_i \ni \mathbb{R}$$

- For two matrices $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times k}$ can define new matrix as:

$$(AB)_{ij} = \sum_{\ell=1}^{n} A_{i\ell} B_{\ell j}$$

- **Exercise:** Determine the number of operations required to compute $A\boldsymbol{v}$ and AB.

Vector norms

Definition

A **norm** is a function $\|\cdot\| : \mathbb{R}^n \to \mathbb{R}_+$ satisfying

1. Positive homogeneity: $\|\alpha \boldsymbol{v}\| = |\alpha| \|\boldsymbol{v}\|$ for all $\alpha \in \mathbb{R}$ and $\boldsymbol{v} \in \mathbb{R}^n$.
2. Definite-ness: $\|\boldsymbol{v}\| = 0$ if and only if $\boldsymbol{v} = \boldsymbol{0}$.
3. Triangle inequality: $\|\boldsymbol{v} + \boldsymbol{w}\| \leq \|\boldsymbol{v}\| + \|\boldsymbol{w}\|$. For all $\boldsymbol{v}, \boldsymbol{w} \in \mathbb{R}^n$.

- **p-norms** For any $1 \leq p < \infty$ the function $\|\boldsymbol{v}\|_p := \left(\sum_{i=1}^n |v_i|^p\right)^{1/p}$ is a norm. *Exercise: show this*
- **max norm** $\|\boldsymbol{v}\|_\infty = \max_i |v_i|$ is also a norm.
  Exercise: show that $\lim_{p\to\infty} \|\boldsymbol{v}\|_p = \|\boldsymbol{v}\|_\infty$
- **Example:** If $\boldsymbol{v} = [1, 3, 7]^\top$ then:

$$\|\boldsymbol{v}\|_2 = \left(1^2 + 3^2 + 7^2\right)^{1/2} = (1 + 9 + 49)^{1/2} = (59)^{1/2} \approx 7.68$$

- **Useful Facts:**

$$(\boldsymbol{v}, \boldsymbol{v}) = \|\boldsymbol{v}\|_2^2$$

$$(\boldsymbol{v}, \boldsymbol{w}) \leq \|\boldsymbol{v}\|_2 \|\boldsymbol{w}\|_2 \quad \text{(Cauchy-Schwarz inequality)}$$

Matrix norms 1: Induced norms

- Vector norms $\|\cdot\|_{p_1}$ on $\mathbb{R}^m$ and $\|\cdot\|_{p_2}$ on $\mathbb{R}^n$ **induce** a norm on $\mathbb{R}^{m\times n}$:

$$\|A\|_{p_2\to p_1} := \sup_{\boldsymbol{v}\neq 0} \frac{\|A\boldsymbol{v}\|_{p_1}}{\|\boldsymbol{v}\|_{p_2}} = \sup_{\|\boldsymbol{v}\|_{p_2}=1} \|A\boldsymbol{v}\|_{p_1}$$

- Will usually write $\|A\|_p$ for $\|A\|_{p\to p}$.
- **Properties:**

$$\|A\boldsymbol{v}\|_{p_1} \leq \|A\|_{p_2\to p_1}\|\boldsymbol{v}\|_{p_2}$$
$$\|AB\|_p \leq \|A\|_p\|B\|_p \quad \text{(submultiplicativity)}$$

- **Special Case 1:** $\|A\|_2 = \lambda_{max}(A^\top A)^{1/2}$ ($\lambda_{max}(\cdot)$ denotes the largest eigenvalue).
- **Special Case 2:** $\|A\|_1 = \max_{1\leq j\leq m} \sum_{i=1}^{n} |a_{ij}|$ *Max. absolute column sum*
- **Special Case 3:** $\|A\|_\infty = \max_{1\leq i\leq n} \sum_{j=1}^{m} |a_{ij}|$ *Max. absolute row sum*

Matrix norms 2: The Frobenius Norm

- The **Frobenius norm** is the sum-of-squares norm:

$$\|A\|_F := \left(\sum_{i,j} a_{ij}^2 \right)^{1/2} = \left(\sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij}^2 \right)^{1/2}$$

- Has an associated **inner product**:

$$(A, B) = \sum_{i,j} a_{ij} b_{ij} = \text{trace}(A^\top B)$$

  Note that (A, b) is a **scalar**.

Example

If $A = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 4 \\ 3 & 6 \end{bmatrix}$ then:

$$(A, B) = (1)(-1) + (3)(4) + (2)(3) + 4(6) = -1 + 12 + 6 + 24 = 41$$

- Note that $(A, A) = \|A\|_F^2$

Linear Independence and Rank

- Vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_r$ are **linearly independent** if no non-trivial linear combination gives zero:

$$\sum_{i=1}^{r} \alpha_i \boldsymbol{v}_i = \mathbf{0} \Leftrightarrow \alpha_1 = \alpha_2 = \ldots = \alpha_r = 0$$

Question: Can you think of other equivalent definitions of linearly independent?

- A has **rank** r if r is the maximum number of linearly independent columns of A.
- **Example** $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 0 & 2 \\ 3 & 1 & 4 \end{bmatrix}$ has rank 2.

Question: Can you think of other equivalent definitions of rank?

- The **outer product** of two vectors is a rank one matrix:

$$\text{if } \boldsymbol{u} = \begin{bmatrix} 1 \\ 3 \end{bmatrix} \text{ and } \boldsymbol{v} = \begin{bmatrix} -2 \\ 4 \end{bmatrix}$$

$$\text{Then: } \boldsymbol{u}\boldsymbol{v}^\top := \begin{bmatrix} 1(-2) & 1(4) \\ 3(-2) & 3(4) \end{bmatrix} = \begin{bmatrix} -2 & 4 \\ -6 & 12 \end{bmatrix}$$